Phase-by-Phase Execution Plan
Long Island Traffic Accident Cluster Discovery in R

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Contents

1 Overview and Design Principles

This document translates the SRS into a concrete, step-by-step execution plan for the Long Island accident clustering project. Each phase has a single, well-scoped objective, a precise list of tasks, explicit done criteria, and notes on known data issues that must be resolved within that phase.

The validated dataset (`li_accidents_clean.csv`) contains **33,375 rows** and **48 columns**. The analysis is split into two complementary pipelines:

1. **Scenario pipeline**: mixed-type accident archetype discovery using `clustMixType::kproto()`
2. **Spatial pipeline**: highway corridor hotspot discovery using `dbscan::dbscan()`

Each pipeline proceeds independently and their results are joined for interpretation only in Phase 9.

1.1 Confirmed Data Issues Requiring Action

A pre-analysis audit of `li_accidents_clean.csv` revealed the following issues that must be resolved during execution:

Variable	Issue	Resolution
`Duration_Minutes`	Max = 2,236,406 min. 3,436 rows (10.3%) above IQR upper fence of 270 min. 6 records exceed one week.	Cap at 99th percentile (≈480 min) and apply log transform. Addressed in Phase 3.
`Distance(mi)`	35.1% of rows have Distance = 0. 251 records exceed 10 miles. Likely a reporting artifact, not a genuine trip distance.	Exclude from the clustering feature matrix. Addressed in Phase 4.
`Wind_Direction`	String duplicates present: "South" and "S", "West" and "W", "North" and "N", "East" and "E" all coexist.	Normalize to 8-direction abbreviation scheme. Addressed in Phase 3.
`Bump`, `Roundabout`, `Turning_Loop`, `Traffic_Calming`, `No_Exit`	All are 0% TRUE across all 33,375 rows. Carry zero signal for clustering.	Drop from all feature matrices. Addressed in Phase 4.
`Give_Way`	Only 0.9% TRUE. Near-zero variance.	Exclude from core feature matrix. Addressed in Phase 4.
`Precipitation(in)`	20.15% missing.	Exclude from core model per SRS. Confirmed in Phase 2.
`Wind_Chill(F)`	15.01% missing and highly correlated with Temperature.	Exclude from all models. Confirmed in Phase 2.

`End_Lat / End_Lng`	39.21% missing.	Exclude; use Start coordinates only. Confirmed in Phase 2.

2 Phase 1: Environment and Reproducibility Setup

2.1 Objective

Establish a reproducible R environment before any data is loaded or transformed, so that all subsequent phases produce identical results across machines and sessions.

2.2 Tasks

1. Install and load required packages. The full list is:
   - `tidyverse` (data manipulation and plotting)
   - `lubridate` (timestamp parsing)
   - `clustMixType` (k-prototypes clustering)
   - `dbscan` (density-based spatial clustering)
   - `cluster` (Gower distance reference and silhouette)
   - `fpc` (cluster stability helpers)
   - `ggplot2` (visualization)
   - `sf` or `geosphere` (coordinate projection for DBSCAN)
   - `factoextra` (optional: validation plots)
2. Set a global random seed at the top of the master script:

   ```
   set.seed(42)
   ```

   All subsequent calls that accept a `set.seed` argument or `nstart` parameter must inherit or re-declare this seed to ensure determinism.
3. Organize the working directory so that:
   - `li_accidents_clean.csv` is the sole data input
   - the R script is named `li_clustering.R` (or `li_clustering.Rmd`)
   - output tables and figures write to a `/output/` subfolder
   - no intermediate `.RData` blobs are committed to the repository
4. Confirm that the R version and package versions are recorded. Add a short `sessionInfo()` call at the bottom of the script or Rmd for reproducibility.

2.3 Done Criteria

- All packages install without errors on the target machine.
- The master script runs from a clean R session without manual interventions.
- The folder structure matches the layout described above.

3 Phase 2: Data Ingestion and Raw Profiling

3.1 Objective

Load the raw dataset, confirm its structure matches expectations from the SRS, and produce a documented profiling report before any transformation is applied. This phase is read-only with respect to the data.

3.2 Tasks

1. Load the CSV:

   ```
   df_raw <- read.csv("li_accidents_clean.csv", stringsAsFactors = FALSE)
   ```

2. Verify dimensions: expect 33,375 rows and 48 columns. Halt if either dimension is unexpected.
3. Confirm the county split:
   - Nassau: 19,390 rows
   - Suffolk: 13,838 rows
   - Queens: 147 rows (western LI edge, retained but not the focus)
4. Confirm the severity distribution:
   - Severity 1: 161
   - Severity 2: 28,357
   - Severity 3: 3,938
   - Severity 4: 919

   Note: Severity 2 accounts for 85% of records. Severity is not a clustering input.
5. Compute and tabulate missingness for every column. Confirm the following critical values:
   - `End_Lat` / `End_Lng`: ≈39.2% missing
   - `Precipitation(in)`: ≈20.1% missing
   - `Wind_Chill(F)`: ≈15.0% missing
   - All other clustering candidates: below 1% missing
6. Confirm that five near-zero binary flags carry no signal:
   - `Bump`, `Roundabout`, `Turning_Loop`, `Traffic_Calming`, `No_Exit`: all 0% TRUE
7. Confirm the `Duration_Minutes` anomaly:
   - Median ≈ 78 min
   - Mean ≈ 449 min (inflated by outliers)
   - Max = 2,236,406 min

- 21 records exceed 1,440 min (one day)
- 6 records exceed 10,080 min (one week)

8. Confirm that `Distance(mi)` has 35.1% zero values and a max of 35.6 miles.
9. Inspect `Wind_Direction` for string duplicates ("South" vs "S", "West" vs "W", etc.).
10. Tabulate `Weather_Condition`: confirm that the top 6–8 labels account for the majority of rows and the long tail contains many sparse labels requiring grouping.
11. Document all findings in a short profiling summary. This summary feeds directly into the data cleaning justifications in the final report.

3.3 Done Criteria

- All dimension, county, severity, and missingness checks pass.
- Known anomalies (Duration outliers, Distance zeros, Wind_Direction duplicates) are confirmed and recorded.
- No data transformations have been applied yet.

4 Phase 3: Feature Engineering

4.1 Objective

Derive analysis-ready features from raw columns. This phase produces all engineered variables that will be used in the clustering input matrices.

4.2 Tasks

3A. Timestamp Features

1. Parse `Start_Time` to a `POSIXct` object using `lubridate::ymd_hms()`.
2. Derive the following variables:
   - `hour`: integer 0–23
   - `hour_band`: factor with levels `Early_AM` (0–5), `AM_Rush` (6–9), `Midday` (10–14), `PM_Rush` (15–18), `Evening` (19–21), `Late_Night` (22–23)
   - `weekday`: factor ordered Monday through Sunday
   - `weekend`: logical TRUE if Saturday or Sunday
   - `month`: integer 1–12
   - `season`: factor with levels `Winter` (Dec–Feb), `Spring` (Mar–May), `Summer` (Jun–Aug), `Fall` (Sep–Nov)
   - `rush_hour`: logical TRUE if `hour` is in 6–9 or 15–18
3. Verify that no `NA` values appear in any derived temporal feature.

3B. Duration Treatment

1. Confirm that the 99th percentile of `Duration_Minutes` is approximately 480 minutes.
2. Create `Duration_cap`: replace any value above 480 with 480. This caps the 1% of extreme outlier records without discarding them.
3. Create `log_Duration`: apply `log1p(Duration_cap)` to compress the right-skewed distribution. This variable is a candidate for the scenario feature matrix.
4. Do not delete the original `Duration_Minutes` column; it is retained for reference.
5. Verify: `summary(log_Duration)` should show no extreme values and a reasonable distribution.

3C. Weather Grouping

1. Count the number of distinct `Weather_Condition` labels (approximately 50–60 in the raw data).
2. Collapse labels into a `weather_group` factor with the following levels:

- `Clear`: "Fair", "Clear", "Fair / Windy"
- `Cloudy`: "Mostly Cloudy", "Partly Cloudy", "Scattered Clouds", "Overcast", "Cloudy", "Mostly Cloudy / Windy", "Partly Cloudy / Windy", "Cloudy / Windy"
- `Rain`: "Light Rain", "Rain", "Heavy Rain", "Light Drizzle", "Light Rain / Windy", "Heavy Rain / Windy", "Drizzle"
- `Snow_Ice`: "Light Snow", "Snow", "Heavy Snow", "Wintry Mix", "Light Freezing Rain", "Ice Pellets", "Sleet"
- `Fog_Haze`: "Fog", "Haze", "Smoke", "Mist", "Patches of Fog", "Shallow Fog"
- `Thunderstorm`: any label containing "Thunder"
- `Other`: all remaining labels with fewer than 30 records

3. Verify the distribution of `weather_group` with a frequency table. The `Clear` and `Cloudy` groups should dominate; `Other` should contain fewer than 200 records total.

3D. Wind Direction Normalization

1. Identify all unique values of `Wind_Direction`. The raw data contains both abbreviated and full-word forms: "South" and "S", "West" and "W", "North" and "N", "East" and "E".
2. Normalize to standard 8-direction abbreviations: N, NE, E, SE, S, SW, W, NW.
3. Map "CALM" and "Variable" / "VAR" to a separate level `Variable_Calm`.
4. Confirm the final `wind_dir_norm` factor has at most 9 levels.
5. This variable is optional for the core scenario model; include it only if the number of factor levels does not dominate the Gower distance calculation. If in doubt, exclude it and note this decision.

3E. Binary Flag Audit

1. Confirm zero-variance flags: `Bump`, `Roundabout`, `Turning_Loop`, `Traffic_Calming`, `No_Exit` are all identically FALSE. Mark these as excluded.
2. Confirm near-zero flags: `Give_Way` (0.9% TRUE), `Station` (0.7% TRUE), `Railway` (0.15% TRUE), `Amenity` (0.34% TRUE). These carry minimal signal. Exclude `Give_Way`, `Station`, `Railway`, and `Amenity` from the core scenario matrix. Document this decision.
3. Retain for the core scenario matrix: `Crossing` (10.1%), `Junction` (16.0%), `Traffic_Signal` (9.5%), `Stop` (1.0%).
4. Convert all retained binary flags to `factor` type with levels `TRUE` and `FALSE` so `kproto()` treats them as categorical.

4.3 Done Criteria

- All engineered features are present in the working data frame with correct types.
- `weather_group`, `hour_band`, `season`, `weekday` are factors.
- `log_Duration` exists and has no extreme values.
- `Wind_Direction` is normalized to at most 9 levels.
- All zero-variance and near-zero-variance flags are marked for exclusion.

5 Phase 4: Build Clustering Input Matrices

5.1 Objective

Construct the final feature matrices for each pipeline. Nothing outside these matrices enters the clustering algorithms.

5.2 Tasks

4A. Scenario Matrix

1. The scenario feature matrix (`df_scenario`) contains exactly the following columns:

   Temporal factors:

   - `hour_band` (factor)
   - `weekday` (factor)
   - `weekend` (factor or logical converted to factor)
   - `season` (factor)
   - `rush_hour` (factor)
   - `Sunrise_Sunset` (factor: Day / Night)

   Weather factors:

   - `weather_group` (factor, 7 levels)

   Numeric weather:

   - `Temperature(F)` (numeric)
   - `Humidity(%)` (numeric)
   - `Pressure(in)` (numeric)
   - `Visibility(mi)` (numeric)
   - `Wind_Speed(mph)` (numeric)

   Road context factors:

   - `Crossing` (factor)
   - `Junction` (factor)
   - `Traffic_Signal` (factor)
   - `Stop` (factor)

   Duration (optional):

   - `log_Duration` (numeric) — include only if less than 5% of values are NA after capping

2. Confirm the following variables are **not** present in `df_scenario`:

   - `Severity`, `ID`, `Source`, `Description`, `Street`, `City`, `Zipcode`, `nzip`

- `Start_Lat`, `Start_Lng`, `End_Lat`, `End_Lng`
- `County` (used for post-cluster comparison only)
- `Distance(mi)` (35% zeros, not a scenario characteristic)
- `Precipitation(in)`, `Wind_Chill(F)` (excessive missingness)
- All zero/near-zero flags identified in Phase 3E

3. Handle remaining missing values in `df_scenario`:
   - For numeric columns with fewer than 1% missing (Temperature, Humidity, Pressure, Visibility, Wind_Speed): impute with the column median. Record how many rows were imputed.
   - For factor columns: if fewer than 10 rows are missing, impute with the mode. If more, treat as a separate `Unknown` level.
4. Confirm the final `df_scenario` has zero missing values before proceeding to Phase 5.
5. Record the final column list and row count in a comment block at the top of the clustering script.

4B. Spatial Matrix

1. The spatial feature matrix (`df_spatial`) contains only:
   - `x_km`: Start_Lng converted to approximate kilometers
   - `y_km`: Start_Lat converted to approximate kilometers
2. Use the flat-earth approximation centered on the LI midpoint ($\approx 40.8°$N, $-73.2°$W):

   ```
   lat0 <- 40.8
   x_km <- (df$Start_Lng - (-73.2)) * 111.32 * cos(lat0 * pi / 180)
   y_km <- (df$Start_Lat - lat0) * 110.574
   ```

   This produces distances in kilometers for the DBSCAN `eps` parameter.
3. Confirm there are no missing values in `Start_Lat` or `Start_Lng`.
4. Confirm the spatial extent of the projected coordinates corresponds to Long Island's east-west span ($\approx$170 km) and north-south span ($\approx$40 km).

5.3 Done Criteria

- `df_scenario` has zero NAs, correct column types, and the documented variable set.
- `df_spatial` has two numeric columns, no NAs, and correct geographic extent.
- Both matrices have 33,375 rows (same row order as the original data for label joining).

6 Phase 5: k-Prototypes Scenario Clustering

6.1 Objective

Run the k-prototypes algorithm on `df_scenario` for candidate values of k, using Gower-style normalization, repeated starts, and a consistent random seed.

6.2 Tasks

1. Verify that all factor columns in `df_scenario` are `factor` type and all numeric columns are `numeric` type. `kproto()` will error if this is not satisfied.

2. Set `lambda = NULL` to let `kproto()` estimate the numeric-to-categorical weight ratio automatically. Document the estimated lambda in the report.

3. Use `type = "gower"` to range-normalize numeric variables before computing distances. This prevents high-variance numeric columns (e.g. Temperature) from dominating the distance calculation.

4. Run `kproto()` for $k \in \{3, 4, 5, 6, 7, 8\}$ with `nstart = 10`. Store all six solutions in a named list:

```
kp_results <- list()
for (k in 3:8) {
  set.seed(42)
  kp_results[[as.character(k)]] <- kproto(
    x     = df_scenario,
    k     = k,
    nstart = 10,
    type  = "gower",
    verbose = FALSE
  )
}
```

5. For each solution, record:

   - within-cluster total distance (the objective function value, `$tot.withinss`)
   - cluster sizes (check for any cluster with fewer than 100 rows)
   - whether any cluster is trivially dominated by a single factor level

6. Plot `tot.withinss` against k to produce an elbow plot. Label each point with its k value.

7. Flag any k solution where the smallest cluster contains fewer than 200 rows as potentially unstable. Do not automatically discard it, but note it for Phase 6 stability checks.

6.3 Done Criteria

- Six candidate `kproto` solutions exist in `kp_results`.
- Elbow plot is produced and saved.
- Cluster sizes are tabulated for all six solutions.
- No solution crashes due to missing values or type mismatches.

7 Phase 6: Cluster Validation and Stability

7.1 Objective

Select the best value of k using internal validation metrics and bootstrap stability, then confirm that the chosen solution is academically defensible.

7.2 Tasks

6A. Internal Validation

1. Run `clustMixType::validation_kproto()` across the candidate k range. This function computes internal indices suited to mixed-type data (including within-cluster compactness relative to between-cluster separation):

```
val_result <- validation_kproto(
  method = "kproto",
  object = df_scenario,
  k      = 3:8,
  nstart = 5,
  type   = "gower"
)
```

2. Inspect and compare the validation indices across all k values. Look for:
   - a clear "knee" in the within-cluster distance
   - consistently better separation scores at a particular k
   - no sharp degradation in balance between k and $k + 1$
3. Do not choose k solely to maximize one metric. Balance interpretability (can each cluster be described meaningfully?) against validation evidence.

6B. Bootstrap Stability

1. Run `clustMixType::stability_kproto()` on the top two candidate k values from 6A:

```
stab <- stability_kproto(
  object = df_scenario,
  k      = k_best,
  B      = 50,
  type   = "gower"
)
```

2. Inspect the average Jaccard similarity for each cluster across bootstrap replicates. A Jaccard above 0.6 for all clusters is considered stable; above 0.75 is strong.

3. If the chosen k has one or more clusters with Jaccard below 0.5, note this as a limitation and prefer the next-best validated k.

4. Record both the validation indices and the Jaccard summary in a table for the report.

6C. Cluster Profile Check

1. For the selected k, run `clustMixType::clprofiles()` to produce factor-level bar plots and numeric boxplots for each cluster.

2. Confirm that at least two clusters have a clearly different dominant weather group, hour band, or roadway context. If all clusters look nearly identical in their profiles, the chosen k is too high or the feature matrix needs revision.

3. Attach plain-English working names to the clusters at this stage. These are preliminary labels that will be refined in Phase 9 after severity and county profiling. Examples:

   - *AM Rush Corridor*: morning hour band, high Junction, mostly Clear weather
   - *Night Low-Visibility*: Night Sunrise_Sunset, Fog or Rain weather group
   - *Weekend Surface Road*: weekend TRUE, Traffic_Signal heavy, Midday or Evening hour band
   - *Winter Adverse*: Snow_Ice weather group, low Temperature, low Visibility

7.3 Done Criteria

- One final value of k is selected and formally justified with validation and stability evidence.
- Each cluster has a working plain-English name.
- A cluster sizes table and the Jaccard summary are ready for the final report.

8 Phase 7: DBSCAN Spatial Hotspot Clustering

8.1 Objective

Identify dense accident corridor clusters along Long Island's major east-west highways using density-based spatial clustering. This pipeline runs entirely on the projected coordinate matrix `df_spatial`.

8.2 Tasks

7A. Parameter Selection

1. Run a k-nearest-neighbor distance plot to choose `eps`:

```
knn_dist <- kNNdist(df_spatial, k = 5)
knn_sorted <- sort(knn_dist)
plot(knn_sorted, type = "l", xlab = "Points sorted by distance",
     ylab = "5-NN distance (km)", main = "kNN Distance Plot")
```

2. Identify the "elbow" in the kNN distance curve. On Long Island's highway-dominated coordinate set, expect the elbow to fall in the range of 0.3–0.8 km given the density of highway segments.
3. Set `minPts` between 10 and 30. A higher `minPts` suppresses small isolated clusters and emphasizes true highway-density zones. Start with `minPts = 15` as the initial candidate.
4. Record the chosen `eps` and `minPts` values and justify them with reference to the kNN distance plot.

7B. Run DBSCAN

1. Run DBSCAN with the selected parameters:

```
db_result <- dbscan(df_spatial, eps = eps_chosen, minPts = minPts_chosen)
```

2. Tabulate the results:
   - number of non-noise clusters (cluster label > 0)
   - size of each cluster
   - percentage of rows labeled as noise (cluster label $= 0$)
3. Accept the solution if the noise percentage is between 5% and 30%. If noise exceeds 50%, `eps` is too small. If there is only one cluster, `eps` is too large. Adjust and re-run if needed, documenting each trial.
4. Inspect cluster shapes by plotting `x_km` vs `y_km` colored by cluster label. Expect elongated east-west shapes corresponding to the LIE (≈latitude 40.77), Southern State Pkwy (≈latitude 40.70), and Northern State Pkwy (≈latitude 40.80).

7C. Corridor Assignment

1. For each non-noise DBSCAN cluster, compute the centroid latitude and longitude. Cross-reference centroid latitude against known corridor latitudes to assign a corridor label:

Centroid Latitude (approx)	**Corridor**
40.74 – 40.76	Long Island Expressway (I-495)
40.68 – 40.72	Southern State Parkway
40.78 – 40.82	Northern State Parkway
40.64 – 40.68	Sunrise Highway
40.72 – 40.76 (north-south elongated)	Meadowbrook / Wantagh Pkwy

2. Clusters that do not align with any known major corridor are labeled as *Surface Road Cluster* or by the dominant city name.

3. Record which major highways appear as DBSCAN clusters and which do not. Note any unexpected hotspot zones.

8.3 Done Criteria

- DBSCAN solution has between 3 and 15 non-noise clusters with noise between 5% and 30%.
- Cluster shapes visually correspond to LI highway corridor alignments.
- Each cluster has a corridor label or a surface-road label.
- Parameter selection is justified with the kNN distance plot.

9 Phase 8: Post-Clustering Interpretation

9.1 Objective

Join both sets of cluster labels back to the original data and produce the interpretation tables and profiles that will drive the final report.

9.2 Tasks

8A. Label Joining

1. Add scenario cluster labels to the original data frame:

```
df_raw$scenario_cluster <- kp_final$cluster
```

2. Add spatial cluster labels:

```
df_raw$spatial_cluster <- db_result$cluster
```

3. Confirm that both label vectors have 33,375 elements and no NAs.

8B. Severity Profiling by Scenario Cluster

1. Compute the percentage of Severity 3 and Severity 4 accidents within each scenario cluster.
2. Produce a summary table:

Cluster	**N**	**Sev1 %**	**Sev2 %**	**Sev3 %**	**Sev4 %**
(fill at runtime)					

3. Flag any cluster with a Severity 4 share more than twice the dataset average (2.75%) as a high-risk archetype.

8C. County Composition by Scenario Cluster

1. Compute the Nassau and Suffolk share within each scenario cluster.
2. Produce a cross-tabulation. Identify whether any cluster is predominantly Nassau or predominantly Suffolk, and interpret this in light of the highway geography.

8D. Weather and Time Profile by Cluster

1. For each scenario cluster, compute:
   - dominant `weather_group` (top label and its share)
   - dominant `hour_band`
   - Day vs Night share (`Sunrise_Sunset`)
   - weekend share
   - mean Temperature, Visibility, and Wind_Speed
2. Produce a compact cluster profile table for the report.

8E. Spatial–Scenario Cross-Tabulation

1. For non-noise DBSCAN records only, cross-tabulate `spatial_cluster` against `scenario_cluster`.
2. Identify whether corridor-specific hotspot zones are dominated by particular scenario archetypes. For example, does the LIE corridor cluster map predominantly to an AM Rush archetype? Does the surface road cluster map to a junction-heavy scenario?
3. This cross-tabulation is a key insight of the two-pipeline design.

8F. Final Cluster Naming

1. Using the full profile from 8B through 8E, finalize the plain-English name for each scenario cluster and each spatial hotspot cluster.
2. Scenario cluster names should reflect the dominant time, weather, and road context. Examples:
   - *AM Rush — Clear Highway*
   - *PM Merge Zone — Congested Commute*
   - *Night Low-Visibility — Fog and Rain*
   - *Winter Adverse — Snow and Ice*
   - *Weekend Surface Road — Signal and Crossing*
3. Spatial hotspot names should reference the corridor. Example: *LIE Westbound Corridor*, *Southern State Pkwy Eastbound Corridor*.

9.3 Done Criteria

- Severity, county, weather, and temporal profile tables are complete.
- Spatial–scenario cross-tabulation is produced.
- All clusters have finalized plain-English names.
- No analytical question from the SRS research questions is left unaddressed.

10 Phase 9: Export and Visualization

10.1 Objective

Produce all report-ready tables, figures, and the final labeled dataset for submission.

10.2 Tasks

1. **Elbow plot**: `tot.withinss` vs k for the k-prototypes solutions (Phase 5).
2. **Validation index plot**: internal validation metrics across $k = 3 : 8$ (Phase 6A).
3. **Stability plot**: Jaccard bootstrap summary for the selected k (Phase 6B).
4. **Cluster profile plot**: `clprofiles()` output for the selected scenario clustering (Phase 6C). One panel per cluster showing factor bar charts and numeric boxplots.
5. **kNN distance plot**: the 5-NN distance curve used to select DBSCAN `eps` (Phase 7A).
6. **Spatial hotspot map**: scatter plot of `x_km` vs `y_km` colored by DBSCAN cluster label (Phase 7B). Noise points shown in grey. Highway corridor labels overlaid.
7. **Severity-by-cluster table**: formatted table for the report showing each scenario cluster's Severity 3+4 share (Phase 8B).
8. **County-by-cluster table**: Nassau vs Suffolk composition by scenario cluster (Phase 8C).
9. **Spatial–scenario cross-tabulation**: heat-table or grouped bar chart (Phase 8E).
10. **Labeled CSV export**: write the final data frame with `scenario_cluster` and `spatial_cluster` columns to `output/li_accidents_clustered.csv`.
11. All figures should be saved at 150 DPI or higher in PNG or PDF format to `output/figures/`.
12. All tables should be saved as CSV to `output/tables/` for easy insertion into the LaTeX report.

10.3 Done Criteria

- All nine figures exist in `output/figures/`.
- All tables exist in `output/tables/`.
- The labeled CSV is exported.
- Every visual is readable at print resolution without axis label truncation.

11 Phase 10: Final Report and Submission

11.1 Objective

Assemble the final report, compile the PDF, and prepare the branch for review.

11.2 Tasks

1. Write the final report in LaTeX. Sections to include:
   - Introduction and motivation (derived from SRS Sections 1–3)
   - Dataset description and data audit findings (Phase 2 summary)
   - Feature engineering decisions (Phase 3, with justification table)
   - Method selection rationale (derived from SRS Section 4)
   - Scenario clustering results: elbow plot, chosen k, validation and stability, cluster profiles
   - Spatial hotspot results: kNN plot, DBSCAN map, corridor assignments
   - Interpretation: severity table, county table, spatial–scenario cross-tabulation
   - Limitations: Duration cap, Precipitation exclusion, Distance exclusion, Severity 2 dominance
   - Conclusion: which clusters are most relevant for safety attention
2. Keep the limitations section honest. In particular:
   - acknowledge that Duration capping at 480 min discards real variability in the top 1% of records
   - acknowledge that Precipitation exclusion (20% missing) may underrepresent rain-related accident patterns
   - acknowledge that spatial hotspot shapes are sensitive to the choice of `eps`
3. Compile the PDF from the LaTeX source and verify all figures render correctly.
4. Commit the following files to the `topic-3-nyc-accident-clustering` branch:
   - `Li_Clustering_SRS.tex` and `Li_Clustering_SRS.pdf`
   - `Li_Clustering_Execution_Plan.tex` and PDF
   - `li_clustering.R` (or `li_clustering.Rmd`)
   - `output/` subfolder with all figures, tables, and the labeled CSV
   - `li_accidents_clean.csv` (if not already committed)
5. Push the branch to the remote repository once the repository owner has added the contributor.

11.3 Done Criteria

- The final report PDF compiles without errors.
- All figures are embedded and readable.
- The branch is clean and the commit message describes the submission state clearly.
- The work addresses all five SRS research questions.

12 Phase Summary Table

#	Phase	Key Output	Done Signal
1	Environment Setup	Clean R project, packages installed, seed set	Script runs from cold start
2	Data Profiling	Missingness table, anomaly confirmation	All expected row/column counts verified
3	Feature Engineering	`hour_band`, `season`, `weather_group`, `log_Duration`, wind normalization	Zero NAs in all engineered features
4	Build Input Matrices	`df_scenario` (no leakage), `df_spatial` (km coords)	Both matrices have 33,375 rows, zero NAs
5	k-Prototypes Clustering	Six candidate solutions, elbow plot	No crashes; cluster sizes tabulated
6	Validation & Stability	Selected k, Jaccard summary, cluster profiles	One defensible solution with working names
7	DBSCAN Spatial	Corridor hotspot clusters, kNN plot, map	3–15 clusters, 5–30% noise, corridor shapes confirmed
8	Interpretation	Severity/county/weather tables, spatial–scenario cross-tab, final names	All SRS research questions answered
9	Export	Figures, tables, labeled CSV	All outputs in `output/` folder
10	Report & Submission	Final LaTeX report, PDF, committed branch	PDF compiles, branch is clean

13 Variable Decision Reference

The following table provides a single-source-of-truth for every variable decision made in this execution plan.

Variable	Decision	Reason
`hour_band`	Include	Key temporal feature, derived from Start_Time
`weekday`	Include	Commuter vs non-commuter pattern signal
`weekend`	Include	Captures Saturday/Sunday off-peak behavior
`season`	Include	Weather-linked seasonal variation
`rush_hour`	Include	Direct commuter proxy
`Sunrise_Sunset`	Include	Day/Night is a strong accident context variable
`weather_group`	Include	Collapsed Weather_Condition; reduces sparsity
`Temperature(F)`	Include	Numeric, no missing, strong season proxy
`Humidity(%)`	Include	Numeric, no missing

Pressure(in)	Include	Numeric; 1.8% mild IQR outliers are physically valid
Visibility(mi)	Include	Key safety variable; low-visibility records are real events
Wind_Speed(mph)	Include	Numeric; 2.4% high outliers are physically plausible
Crossing	Include	10.1% TRUE; meaningful road context
Junction	Include	16.0% TRUE; strongest road context flag
Traffic_Signal	Include	9.5% TRUE; distinguishes surface roads from highways
Stop	Include	1.0% TRUE; low but non-trivial; kept as marginal signal
log_Duration	Include (conditional)	After cap at 480 min; carries incident severity signal
Precipitation(in)	Exclude	20.1% missing; too high for routine imputation
Wind_Chill(F)	Exclude	15.0% missing; redundant with Temperature
Distance(mi)	Exclude	35.1% zeros; reporting artifact, not scenario signal
Amenity	Exclude	0.34% TRUE; near-zero variance
Give_Way	Exclude	0.9% TRUE; near-zero variance
Station	Exclude	0.7% TRUE; near-zero variance
Railway	Exclude	0.15% TRUE; near-zero variance
Bump	Exclude	0.0% TRUE; zero variance
Roundabout	Exclude	0.0% TRUE; zero variance
Turning_Loop	Exclude	0.0% TRUE; zero variance
Traffic_Calming	Exclude	0.0% TRUE; zero variance
No_Exit	Exclude	0.0% TRUE; zero variance
Severity	Post-cluster only	Leakage risk; profile after clustering
County	Post-cluster only	Use for Nassau/Suffolk comparison after clustering
Start_Lat / Start_Lng	Spatial only	Used only in df_spatial for DBSCAN
End_Lat / End_Lng	Exclude	39.2% missing
ID, Source, Description, Street, City, Zipcode, nzip	Exclude	Identifiers and free text; no clustering value